Text-RSSR: Text-Conditioned Remote Sensing Super-Resolution with Cross-Modal Learning

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***Abstract*—Super-resolution of remote sensing images (RSSR) seeks to reconstruct high-resolution content from low-resolution inputs, thereby providing critical detail for tasks such as environmental monitoring, urban mapping, and land-cover analysis. While deep learning-based RSSR methods have recently advanced super-resolution performance, they remain limited in effectively capturing semantic context. Most existing approaches rely solely on visual information, which often leads to over-smoothed textures, inadequate structural detail, and poor generalization across diverse landscapes. To address these challenges, this paper proposes Text-RSSR, a text-conditioned RSSR framework that introduces textual and semantic guidance through cross-modal learning. By leveraging text captions and CLIP embeddings, the model incorporates high-level contextual priors to iteratively refine visual features and enhance reconstruction quality. This integration of linguistic and visual information enables the network to achieve a stronger balance between fine detail restoration and semantic consistency. The effectiveness of Text-RSSR is validated through extensive experiments on three public RSSR datasets, achieving state-of-the-art performance with PSNR values of 16.3632 dB on the Alsat-2B dataset, 27.4904 dB on the UC Merced Land Use Dataset, and 27.4089 dB on the Aerial Image Dataset. In addition, we evaluate Text-RSSR on the ImageNet Large Scale Visual Recognition Challenge 2012 dataset, where it also demonstrates the best performance with a PSNR of 1 dB. The implementation will be made publicly available at https://github.com/haoyangofficial/textrssr.**

***Index Terms*—Remote sensing image, super-resolution, multi-modal learning**

I. Introduction

HIGH-RESOLUTION (HR) remote sensing (RS) imagery underpins a broad spectrum of critical applications, ranging from environmental monitoring [1] and urban planning [2] to precision agriculture [3], disaster response [4], and national security. In these domains, fine-grained structural and spectral detail is often indispensable for reliable land-cover classification, object detection, and temporal change analysis [5]. However, acquiring high-resolution RS imagery is frequently constrained by hardware limitations, high acquisition costs, atmospheric perturbations, and limited temporal coverage [6]. This reality has motivated a substantial body of research into methods that computationally enhance image resolution, reconstructing HR data from low-resolution (LR) inputs to bridge the gap left by physical constraints.

Early efforts in Remote Sensing Super-Resolution (RSSR) employed classical interpolation techniques, gradually giving way to convolutional neural network (CNN)-based models that directly learn an end-to-end mapping from LR to HR images [7]. For instance, CNNs trained for natural image super-resolution have been successfully adapted to RS tasks [8]. Subsequent advancements introduced generative adversarial networks (GANs), such as SRGAN [9], which enhanced perceptual quality by incorporating adversarial loss along with standard pixel-level objectives. RS-specific refinements followed: NDSRGAN focused on dense sampling in real aerial imagery [10], and Dual RSS-GAN combined multi-source satellite data for improved remote sensing super-resolution [11]. More recently, transformer-enhanced architectures that fuse multi-scale receptive fields with hybrid self-attention, such as multi-scale convolution plus hybrid Transformer model, have demonstrated significant gains in PSNR, SSIM, and feature similarity over SRGAN baselines [12]. Complementing these advances, a comprehensive review of RSSR research highlights growing methodological diversity spanning CNNs, GANs, attention mechanisms, and evaluation benchmarks [13].

Recent advances in cross-modal learning and vision–language models (VLMs) have demonstrated the potential of integrating textual and visual modalities to achieve richer and more semantically grounded representations [14]. Textual descriptions can encapsulate high-level contextual knowledge, such as “a dense urban area with high-rise buildings,” or “a coastal region with sandy beaches.” Incorporating such semantic priors into SR reconstruction has the potential to guide the model toward producing more realistic and semantically consistent HR imagery. A unique benefit of text-guided SR is that text captions of HR images can preserve the high-frequency details of the original image before degradation while requiring negligible storage, making them a cost-effective auxiliary modality [15]. In natural image domains, text-conditioned generation and reconstruction have already shown promise in enhancing both perceptual quality and interpretability [16]. However, in the context of RS, existing cross-modal studies have largely concentrated on tasks such as scene classification, image–text retrieval, and captioning [17]. These works highlight the feasibility of aligning visual and linguistic representations in the RS domain, but no prior research has systematically investigated textual conditioning for super-resolution. This represents a critical gap, as SR performance could be significantly improved by leveraging auxiliary semantic supervision from language, thereby enhancing generalization and ensuring

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greater semantic fidelity.

To fill this gap, this study proposes Text-RSSR, a novel framework for text-conditioned remote sensing super-resolution leveraging cross-modal learning. The Text-RSSR framework comprises four main steps: Data Preparation, Multimodal Encoding, Text-Guided Iterative Super-Resolution, and Image Decoding. In the Data Preparation stage, a VLM generates a text caption from the HR image, capturing its detailed content. During Multimodal Encoding, the LR image is processed through both a CNN Image Encoder and a CLIP Image Encoder to obtain Visual Embeddings, while the text caption is encoded by a CLIP Text Encoder to produce Text Embeddings. In the CLIP-Guided Iterative Super-Resolution stage, the CNN Visual Embeddings are progressively refined and aligned with textual information and semantic guidance. At last, in Image Decoding, the refined Visual Embeddings are decoded by a CNN decoder and upsampled to produce the final high-resolution image. The key contributions of this article include:

1) This work proposes Text-RSSR, a text-conditioned RSSR approach based on cross-modal learning. By leveraging textual cues and semantic guidance, the model iteratively reconstructs RS images with a deeper and more comprehensive understanding informed by multiple modalities.
2) This work introduces a Text-Guided Iterative Super-Resolution process that progressively enhances LR images by integrating textual cues (Text Embeddings captured by the CLIP Text Encoder) and semantic information (CLIP Visual Embeddings), thereby enabling finer detail recovery and stronger alignment with textual descriptions.
3) This study develops a dual-head network that captures fine-grained local visual features from images while simultaneously maintaining the global textual and semantic representations provided by CLIP.

The remainder of this paper is organized as follows: Section II reviews related work in RSSR and multimodal learning in RSSR. Section III describes the proposed Text-RSSR framework in detail. Section IV outlines implementation details, experimental datasets, results and ablation studies. Finally, Section V concludes this paper.

II. Related Works

A. Remote Sensing Super-Resolution Methods

RSSR aims to reconstruct HR images from LR observations, thereby enhancing spatial details crucial for tasks such as land cover classification, urban monitoring, and environmental analysis. Classical RSSR approaches primarily rely on signal processing or statistical modeling. Interpolation-based techniques, such as bicubic interpolation [18], provide simple baselines but often result in blurry reconstructions due to their inability to recover high-frequency details. Sparse representation methods, such as sparse coding-based SR [19], improve over interpolation by learning dictionaries of high- and low-resolution patches, but they are still limited in capturing the complex spatial and spectral structures inherent in remote sensing images.

In recent years, deep learning (DL)-based methods have achieved remarkable progress in RSSR by learning nonlinear mappings from LR to HR images. Convolutional neural networks (CNNs), such as SRCNN [20] and EDSR [21], pioneered single-image RSSR by directly learning end-to-end super-resolution functions. More recently, state-space or Mamba-based architectures have emerged, such as FMSR [22] and EMAN [23], which model long-range dependencies efficiently. GANs, including SRGAN [9] and CDGAN [24], have been widely adopted to generate perceptually realistic HR images. Diffusion-based models, such as SRDiff [25] and DDRM [26], leverage probabilistic generative frameworks to progressively refine image quality. Despite these advances, existing DL-based methods still face significant limitations: they primarily focus on unimodal visual information and rarely leverage auxiliary modalities, particularly textual descriptions, which can provide high-level semantic context about the scene.

To address this limitation, this study introduces Text-Conditioned Remote Sensing Super-Resolution (Text-RSSR), a multimodal framework designed to incorporate textual information into the RSSR process. Text-RSSR leverages VLMs to generate captions from HR images, which are then encoded into embeddings using CLIP [14]. In this framework, CNN visual embeddings serve as the primary modality for reconstruction, while textual cues and semantic guidance act as auxiliary signals to refine and enhance feature learning [27]. A Text-Guided Feature Alignment module progressively refines CNN visual embeddings with textual information and semantic guidance before decoding them into the final HR image. By integrating textual information and semantic guidance, Text-RSSR aims to overcome the semantic gap of prior unimodal methods, offering a novel pathway toward semantically enriched and context-aware RSSR.

B. Multimodal Learning in Remote Sensing Super-Resolution

Multimodal Learning in RSSR refers to approaches that leverage complementary data types, such as Synthetic Aperture Radar (SAR), optical imagery, hyperspectral and multispectral images, or even textual/contextual information, to enhance the spatial and/or spectral fidelity of reconstructed HR images.

Multimodal learning-based RSSR methods can broadly be divided into cross-modal fusion methods and joint embedding or contrastive learning methods. Cross-modal fusion approaches seek to integrate complementary information from multiple sensors to enhance reconstruction quality. For example, the Multi-Resolution Collaborative Fusion of SAR, Multispectral and Hyperspectral Images method combines SAR, multispectral, and hyperspectral imagery through deep fusion strategies to preserve spatial, spectral, and geometrical fidelity in reconstructed outputs, demonstrating effective cross-modal integration in RSSR contexts [28]. Another example involves combining LR hyperspectral with HR multispectral data through transformer-based fusion or retractable spatial–spectral modeling, yielding better spectral and spatial details [29]. In contrast, joint embedding and contrastive learning methods focus on learning a shared latent representation across

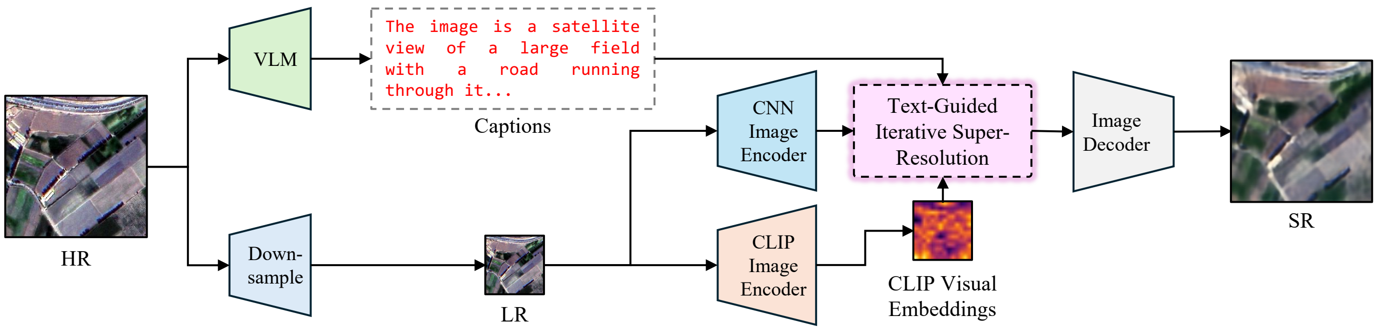


Fig. 1. Overall architecture of the purposed Text-Conditioned Remote Sensing Super-Resolution Network (Text-RSSR).

modalities. One representative example is the multimodal contrastive learning framework for RS tasks, which employs a dual-encoder architecture pre-trained on approximately one million paired Sentinel-1 (radar) and Sentinel-2 (optical) image pairs. This framework aligns features across modalities via contrastive learning and shows improved performance in downstream tasks like segmentation and land-cover mapping [30]. Another compelling approach is the Contrastive Radar-Optical Masked Autoencoders, which fuses cross-modal contrastive learning with masked autoencoding to learn rich multimodal representations from spatially aligned SAR and multispectral optical imagery [31]. Despite the promise of these fusion and embedding-based RSSR approaches, they tend to rely heavily on additional visual modalities (such as SAR, hyperspectral, or multispectral data), while largely neglecting the potential of textual information, including captions and semantic descriptions, that could provide high-level guidance for more semantically aware super-resolution.

To address this gap, this study introduces Text-RSSR, a Text-Conditioned RSSR Network. Text-RSSR synergizes image data with auxiliary textual and semantic cues by first generating captions for HR images, downsampling them, and encoding the LR images via CLIP embeddings. These Text Embeddings and CLIP Visual Embeddings act as complementary guidance, while the CNN Visual Embeddings, serving as the primary modality, undergo an iterative Text-Guided Feature Aligning process within an SR Unit. Through successive refinement steps, the framework effectively fuses the main visual features with auxiliary textual cues and semantic guidance, culminating in a CNN-based decoder that produces enhanced high-resolution outputs. By explicitly incorporating these auxiliary modalities, Text-RSSR remedies the underutilization of language in multimodal RSSR and enables more semantically faithful reconstruction. The overall architecture is shown in Figure 1. While no prior RSSR method explicitly employs text conditioning, related work in natural image SR demonstrates the viability of CLIP-based text guidance, as in the recent Text-Guided Explorable Image Super-Resolution framework [15]. Text-RSSR thus represents a novel extension of multimodal learning to RSSR, integrating visual and linguistic modalities for the first time.

III. Proposed Method

A. Overview

This study introduces a Text-Conditioned Remote Sensing Super-Resolution Network (Text-RSSR) designed to improve remote sensing super-resolution performance through the iterative use of text captions and CLIP embeddings. As shown in Figure 2, the Text-RSSR framework comprises four main steps: Data Preparation, Multimodal Encoding, Text-Guided Iterative Super-Resolution, and Image Decoding.

In the Data Preparation step, a VLM generates a text caption from the high-resolution (HR) image, capturing its detailed content. During Multimodal Encoding, the low-resolution (LR) image is processed by two separate encoders (a CNN Image Encoder and a CLIP Image Encoder) to produce CNN Visual Embeddings and CLIP Visual Embeddings, respectively. Simultaneously, the text caption is processed through a CLIP Text Encoder to generate Text Embeddings. In the CLIP-Guided Iterative Super-Resolution step, the CNN Visual Embeddings are iteratively aligned and refined. In each iteration, all embeddings are input into a Text-Guided Feature Aligning step, followed by processing through a Super-Resolution (SR) Unit. After the final iteration, the refined CNN Visual Embeddings go through one last Text-Guided Feature Aligning. Finally, in the Image Decoding step, the processed CNN Visual Embeddings are decoded by a CNN decoder, upsampled, and further decoded by an additional CNN.

B. Data Preparation

In the Data Preparation step, the HR image captured by the satellite is first downsampled to produce an LR image, which serves as the input for super-resolution. At the same time, the HR image is used to generate a descriptive text caption that summarizes its semantic content, such as terrain type, objects, or structural patterns. This caption is created by a VLM running onboard the satellite, allowing semantic information to be extracted before data transmission.

Since transmitting full HR images requires considerable bandwidth and storage, only the LR image and the lightweight text caption are sent to Earth. This design not only reduces communication cost but also ensures that semantic details are preserved through the caption, compensating for information lost during downsampling. As a result, the prepared data

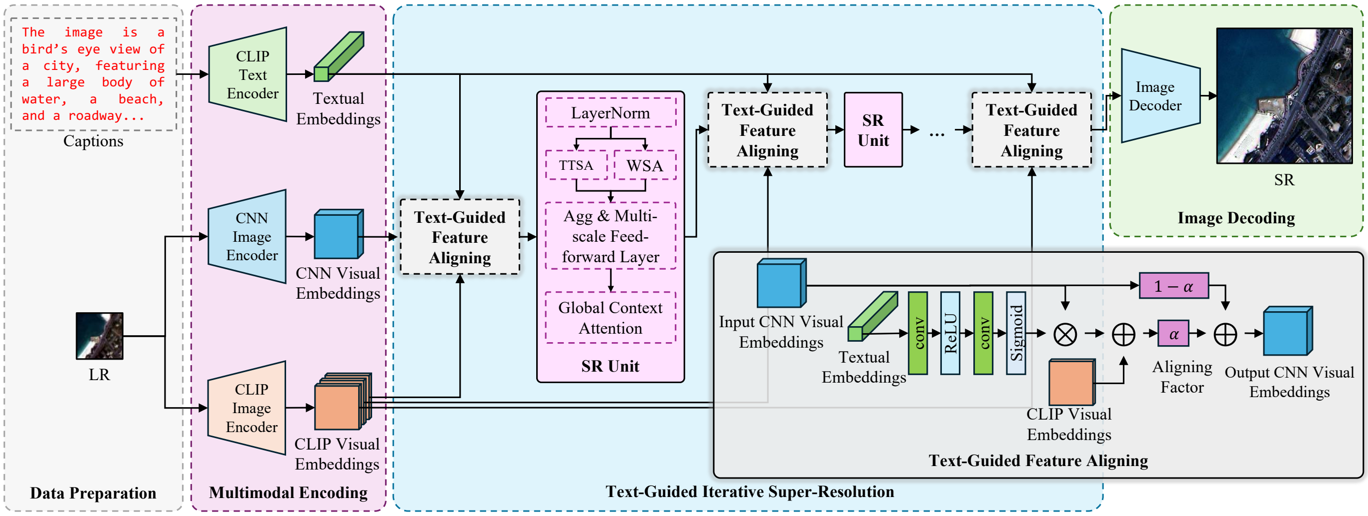


Fig. 2. Overview of the purposed Text-Conditioned Remote Sensing Super-Resolution Network (Text-RSSR). TTSA denotes Top-k Token Selective Attention, and WSA represents Window-based Self-Attention within the TTST architecture.

provide both spatial structure from the LR image and contextual knowledge from the caption, forming the foundation for multimodal super-resolution.

C. Multimodal Encoding

In this process, the LR image is processed by both a CNN Image Encoder and a CLIP Image Encoder to generate CNN Visual Embeddings and CLIP Visual Embeddings, respectively. At the same time, the text caption is processed through a CLIP Text Encoder to produce Text Embeddings.

As discussed, remote sensing images often contain repetitive patterns and similar scene types, making it difficult for the CLIP Visual Encoder alone to effectively distinguish between them. To address this, a dual image encoder architecture is adopted to combine the semantic understanding capability of the CLIP Visual Encoder with the local feature extraction strength of the CNN. In this architecture, the CNN Visual Embeddings serve as the primary features, while the CLIP Visual Embeddings provide supplementary information to guide feature adjustment and aligning.

The image encoding process is described as follows. Let $\mathbf{L} \in \mathbb{R}^{C\times H\times W}$ represent the LR image received from the satellite. The extraction of CNN Visual Embeddings using the CNN Visual Encoder is defined by:

$$\mathbf{X}_0 = \mathrm{VE}_{\mathrm{CNN}}(\mathbf{L}), \tag{1}$$

where $\mathrm{VE}_{\mathrm{CNN}}$ denotes the CNN Visual Encoder, and $\mathbf{X}_0$ represents the resulting CNN Visual Embeddings.

To provide CLIP-based semantic guidance, the first token and the remaining tokens from the CLIP Visual Encoder output are separated. A Semantic Guiding (SG) process is then applied to generate a semantic guidance map for each Text-Guided Feature Aligning step in the Text-Guided Iterative Super-Resolution process [27]. The extraction of CLIP Visual Embeddings and the computation of the semantic guidance map for the i-th Text-Guided Feature Aligning step are defined as follows:

$$\mathbf{S}_i = \mathrm{SG}_i(\mathbf{v}_0, \mathbf{v}_{\mathrm{rm}}), \tag{2}$$

where

$$\mathbf{v}_0, \mathbf{v}_{\mathrm{rm}} = \mathrm{VE}_{\mathrm{CLIP}}(\mathbf{L}).$$

Here, $\mathbf{S}_i$ and SG_i denote the CLIP Visual Embeddings and the Semantic Guiding process for the i-th Text-Guided Feature Aligning step, respectively. $\mathrm{VE}_{\mathrm{CLIP}}$ denotes the CLIP Visual Encoder, $\mathbf{v}_0$ represents the first token of the resulting CLIP Visual Embeddings, and $\mathbf{v}_{\mathrm{rm}}$ denotes the remaining tokens. Let the number of SR Units be l. Then, the number of Text-Guided Feature Aligning steps is $l+1$, with the index $i \in [0, l]$.

The SG process is designed to modulate local feature enhancement in the Super-Resolution pipeline using semantic priors from CLIP. For each step $i \in [0, l]$, the SG process computes a semantic guidance map $\mathbf{S}_i \in \mathbb{R}^{B\times H\times W}$ for each of the l SR Units, providing explicit spatially-aware guidance aligned with the target semantics.

Let $\mathbf{v}_0 \in \mathbb{R}^{B\times C}$ denote the global semantic embedding (i.e., the [CLS] token) and $\mathbf{v}_{\mathrm{rm}} \in \mathbb{R}^{B\times(H\times W)\times C}$ the local semantic embeddings extracted from the CLIP Visual Encoder. The SG process for the i-th step is defined as:

$$\mathrm{SG}_i(\mathbf{v}_0, \mathbf{v}_{\mathrm{rm}}) = \mathrm{CS}(\Theta(\mathbf{e}_i), \Theta(\mathbf{v}_{\mathrm{rm}})), \tag{3}$$

where $\mathrm{CS}(\cdot,\cdot)$ denotes the cosine similarity, and $\Theta(\cdot)$ is the ℓ_2-normalization function applied along the feature dimension. The latent embeddings $\mathbf{e}_i \in \mathbb{R}^{B\times l\times C}$ used for computing the semantic guidance maps are derived through the following stages:

$$\begin{cases} \mathbf{p}_{\mathrm{a}} = \mathrm{MetaNet}(\mathbf{v}_0, \mathbf{p}_0), \\ [\mathbf{p}_{\mathrm{a}}^*; \mathbf{p}_{\mathrm{b}}^*] = \mathrm{MHSA}([\mathbf{p}_{\mathrm{a}}; \mathbf{p}_{\mathrm{b}}]), \\ [\mathbf{e}_i] = \mathrm{MLP}_{\mathbf{e}}(\mathrm{MLP}_{\mathrm{a}}(\mathbf{p}_{\mathrm{a}}^*) \odot \sigma(\mathrm{MLP}_{\mathrm{b}}(\mathbf{p}_{\mathrm{b}}^*))). \end{cases} \tag{4}$$

Here, $\mathbf{p}_0 \in \mathbb{R}^{1\times C}$ is a learnable global prior descriptor, and $\mathbf{p}_{\mathrm{b}} \in \mathbb{R}^{(l+1)\times C}$ are learnable local unit descriptors, shared across all steps. MetaNet is an architecture to extract semantic correspondence between $\mathbf{v}_0$ and $\mathbf{p}_0$ [32]. The Multi-Head Self-Attention (MHSA) block refines the concatenated descriptors $[\mathbf{p}_{\mathrm{a}}; \mathbf{p}_{\mathrm{b}}]$, enabling context exchange between global and unit tokens [33]. The gating mechanism modulates the unit embeddings via element-wise product between the transformed global vector and a gated local vector. The resulting fused descriptors $\mathbf{e}_i$ are compared with the normalized local features $\mathbf{v}_{\text{rm}}$ via cosine similarity, and $i \in [0, l]$.

The output of the semantic guiding process is a semantic guidance map $\mathbf{S}_i \in \mathbb{R}^{B \times H \times W}$, where $H \times W$ corresponds to the resolution of the LR image. This map serves as a soft spatial prior for the i-th Text-Guided Feature Aligning step, guiding the model to enhance semantically meaningful regions in accordance with the text prompt. This design allows each SR Unit to adaptively incorporate global semantic context and local alignment cues, thereby improving spatial correspondence and preserving high-frequency details during the Super-Resolution reconstruction process.

The Text Embeddings are generated using the CLIP Text Encoder as follows:

$$\mathbf{T} = \text{TE}(\mathbf{t}), \tag{5}$$

where $\mathbf{t}$ denotes the caption of the HR image generated by the VLM, TE denotes Text Encoder, and $\mathbf{T}$ represents the resulting Text Embeddings.

D. Text-Guided Iterative Super-Resolution

In this step, the CNN Visual Embeddings are iteratively aligned and refined. To maintain the supervisory influence of the CLIP Visual Embeddings and Textual Embeddings throughout the entire Super-Resolution process, each iteration begins by using these embeddings to align the CNN Visual Embeddings. The aligned CNN Visual Embeddings are then processed through an SR Unit. After the final iteration, the refined CNN Visual Embeddings undergo one final Text-Guided Feature Aligning step. Let the CNN Visual Embeddings input for the i-th iteration be $\mathbf{X}_i$, where $i \in [0, l)$. Then, the i-th iteration is defined as:

$$\mathbf{X}_{i+1} = \text{SR}(\text{TGFA}(\mathbf{X}_i)), \tag{6}$$

where TGFA denotes the Text-Guided Feature Aligning step, and SR denotes the Super-Resolution Unit.

Text-Guided Feature Aligning is designed to align and refine the CNN Visual Embeddings using both the CLIP Visual Embeddings and the CLIP Textual Embeddings. In this process, textual information provides channel-wise attention, enabling the model to identify and emphasize the most relevant feature channels for the Super-Resolution task. Following this channel-wise refinement, the semantic guidance maps are applied to perform pixel-wise refinement, offering direct high-frequency guidance to enhance spatial detail in the reconstructed image.

Let $\mathbf{T} \in \mathbb{R}^d$ be the Textual Embeddings, $\mathbf{S}_i$ the semantic guidance map for the i-th iteration, and d the dimension of CLIP's hidden state. The Text-Guided Feature Aligning process is defined as:

$$\text{TGFA}(\mathbf{X}_i) = \alpha \cdot \text{Align}(\mathbf{X}_i) + (1 - \alpha) \cdot \mathbf{X}_i, \tag{7}$$

where

$$\text{Align}(\mathbf{X}_i) = \mathbf{X}_i \odot \text{TG}(\mathbf{T}) + \mathbf{S}_i.$$

Here, α is a learnable aligning factor that controls the proportion of alignment, $\text{Align}(\mathbf{X}_i)$ denotes the aligned visual embeddings, and $\odot$ represents element-wise multiplication. The function TG is a lightweight MLP composed of a convolution layer, a ReLU activation, another convolution layer, and a Sigmoid activation, projecting the Textual Embeddings into the channel space.

The complete process of Text-Guided Feature Aligning is depicted in the shaded gray area of Figure 2. For the Super-Resolution Unit, we adopt the Residual Token Selective Group (RTSG) from Top-k Token Selective Transformer (TTST) [34] architecture.

E. Image Decoding

In the Image Decoding step, the processed CNN Visual Embeddings are used to generate the super-resolved (SR) image, which serves as the final output of the entire network. The CNN Visual Embeddings obtained from the last Text-Guided Feature Aligning are first passed through a CNN, then upscaled, and subsequently processed by another CNN to produce the SR image.

IV. Experiments

A. Implementation details

This study is implemented using PyTorch [35], and all experiments are conducted on a system equipped with an Intel Core i9-10900X CPU, 128 GiB of RAM, and a single NVIDIA GeForce RTX 4090 GPU. The operating system is Ubuntu 22.04 (Jammy) with CUDA version 12.4. The Text-RSSR framework, along with all baseline models used for comparison, adopts L1 Loss [36] as the loss function and employs the AdamW optimizer [37] with an initial learning rate of 10^{-4}. The learning rate is reduced by half at the midpoint of the training process. The downscale factor used to generate the LR image is set to 4. In this study, the training epochs were set to a moderate number to simulate a computation-constrained scenario. All models are trained from scratch, except for the CLIP components. For Text-RSSR, the VLM used to generate captions is LLaVa-1.5 7B [38]. The CLIP Visual Encoder and Textual Encoder used in the Multimodal Encoding step are pretrained and kept frozen during training. The initial aligning factor α in the Text-Guided Feature Aligning step is set to 0.5. The number of SR Units, l, is set to 6. The RTSG Super-Resolution Unit is initialized following the configuration provided in the original TTST paper [34]. Benchmark evaluations are conducted on three public RSSR datasets: Alsat-2B dataset [39], UC Merced Land Use Dataset [40], and Aerial Image Dataset (AID) [41]; as well as on a common object dataset, the ImageNet Large Scale Visual Recognition Challenge 2012 (ILSVRC2012) dataset [42]. Model performance is assessed using Peak Signal-to-Noise Ratio (PSNR, calculated on the RGB channels) [43] and Structural Similarity Index Measure (SSIM) [44]. Compared to existing methods, Text-RSSR demonstrates strong performance on RSSR tasks.

TABLE I
QUANTITATIVE RESULTS OF THE EXPERIMENT ON ALSAT-2B DATASET. THE BEST RESULT IS HIGHLIGHTED IN **BOLD**, WHILE THE SECOND-BEST RESULT IS UNDERLINED.

Model	Parameters (M)	FLOPs (G)	Average		Agriculture		Special		Urban	
			PSNR (dB)	SSIM	PSNR (dB)	SSIM	PSNR (dB)	SSIM	PSNR (dB)	SSIM
FAT [45]	10.19	43.86	16.0099	**0.3707**	17.5152	**0.4409**	16.6609	0.3946	14.8890	0.3261
FMSR [22]	7.05	30.71	16.2615	0.3539	17.6293	0.4184	16.8002	0.3768	15.3054	0.3117
GCRDN [46]	31.83	443.80	15.8084	0.2875	17.1307	0.3931	16.3803	0.3188	14.8237	0.2258
TSFNet [47]	3.13	111.17	**16.4214**	0.3704	17.7688	0.4376	**17.0326**	**0.3951**	15.3846	0.3255
TTST [34]	18.30	76.84	16.2754	0.3542	17.6870	0.4220	16.8012	0.3764	15.3243	0.3121
Text-RSSR	22.72	76.86	16.3632	0.3702	**18.0134**	0.4385	16.8149	0.3868	**15.4435**	**0.3346**

TABLE II
QUANTITATIVE RESULTS OF THE EXPERIMENT ON UC MERCED LAND USE DATASET. THE BEST RESULT IS HIGHLIGHTED IN **BOLD**, WHILE THE SECOND-BEST RESULT IS UNDERLINED. THE UNIT FOR PSNR IS DECIBELS (DB).

Class	FAT [45]		FMSR [22]		GCRDN [46]		TSFNet [47]		TTST [34]		Text-RSSR	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
Agriculture	**24.9965**	**0.4805**	24.9226	0.4757	24.7018	0.4128	24.5531	0.4259	24.2153	0.4154	24.4920	0.4551
Airplane	27.8155	0.8212	28.2478	0.8320	28.2134	0.8321	28.0971	0.8279	28.4077	0.8343	**28.4183**	**0.8350**
Baseball Diamond	32.9822	0.8533	33.3289	0.8617	33.2192	0.8570	33.2576	0.8595	33.4316	0.8626	**33.4700**	**0.8637**
Beach	36.4693	0.8987	36.7805	0.9061	36.9861	0.9072	37.0140	0.9069	37.0934	0.9082	**37.1166**	**0.9087**
Buildings	23.6873	0.7487	24.1050	0.7656	24.2573	0.7681	23.8823	0.7552	24.3830	0.7737	**24.4572**	**0.7772**
Chaparral	25.3234	0.7543	25.6270	0.7621	25.6057	0.7607	25.6488	0.7598	**25.7485**	0.7672	25.7471	**0.7676**
Dense Residential	25.7298	0.7895	26.2259	0.8049	26.2596	0.8058	26.0135	0.7979	26.4440	0.8106	**26.5231**	**0.8135**
Forest	25.6720	0.6391	25.9337	0.6409	25.9508	0.6401	25.9675	0.6388	25.9932	0.6451	**26.0015**	**0.6463**
Freeway	27.4274	0.7525	27.7678	0.7644	27.8962	0.7647	27.5942	0.7551	27.9259	0.7685	**27.9556**	**0.7711**
Golf Course	33.5669	0.8554	33.8471	0.8625	33.9097	0.8624	33.8675	0.8617	**33.9689**	0.8640	33.9649	**0.8643**
Harbor	21.0594	0.8232	21.5372	0.8368	21.5931	0.8392	21.1637	0.8282	21.8485	0.8495	**21.9199**	**0.8504**
Intersection	25.4345	0.7533	25.8930	0.7690	25.9169	0.7680	25.7135	0.7567	26.0866	0.7759	**26.1001**	**0.7777**
Medium Residential	27.5041	0.8053	28.1087	0.8196	28.1493	0.8205	28.0127	0.8167	28.3415	0.8242	**28.4243**	**0.8262**
Mobile Home Park	17.4454	0.5973	17.6357	0.6131	17.7582	0.6147	17.4915	0.5976	17.8737	**0.6281**	**17.9357**	0.6276
Overpass	24.6758	0.6933	24.9872	0.7054	25.0903	0.7070	24.7842	0.6948	25.0561	0.7091	**25.1763**	**0.7132**
Parking Lot	21.1731	0.7508	21.5318	0.7631	21.5031	0.7605	21.1645	0.7452	21.9376	0.7750	**21.9953**	**0.7788**
River	28.9958	0.7852	29.4090	0.7941	29.4689	0.7959	29.4365	0.7931	29.4905	0.7971	**29.5195**	**0.7978**
Runway	30.3700	0.7884	30.5751	0.7929	30.5688	0.7899	30.4315	0.7849	30.6446	0.7945	**30.8615**	**0.8001**
Sparse Residential	27.2791	0.7770	27.5781	0.7851	27.6738	0.7873	27.5750	0.7831	**27.7425**	**0.7904**	27.7201	0.7891
Storage Tanks	29.3186	0.8208	29.7391	0.8316	29.7951	0.8330	29.7103	0.8287	29.8461	0.8339	**29.8824**	**0.8358**
Tennis Court	28.9392	0.8071	29.3549	0.8217	29.3263	0.8168	29.1162	0.8065	29.4137	0.8227	**29.6164**	**0.8293**
Average	26.9460	0.7617	27.2922	0.7718	27.3259	0.7688	27.1664	0.7631	27.4235	0.7738	**27.4904**	**0.7775**

B. Dataset Description

We conducted experiments on three public RSSR datasets: Alsat-2B dataset [39], UC Merced Land Use Dataset [40], and AID [41]. To further evaluate the super-resolution performance of Text-RSSR on common objects, we conducted additional experiments using the ILSVRC2012 dataset [42]. According to the results, Text-RSSR demonstrates strong performance across all evaluated datasets.

The Alsat-2B dataset, introduced by A. Djerida *et al.* [39], includes 2,182 training samples and three testing subsets (agriculture, urban, and special scenes), containing 56, 282, and 239 image pairs, respectively. Each HR image has a fixed resolution of 256×256, while each LR image has a resolution of 64×64. The spatial resolution of the dataset is 2.5 meters. The Alsat-2B dataset exhibits noticeable color style differences between HR and LR images, making the RSSR task more challenging. All models are trained for 50 epochs, and the batch size for Text-RSSR is set to 4.

The UC Merced Land Use Dataset, introduced by Y. Yang *et al.* [40], consists of 21 categories, each with 100 HR images. Each image has an approximate resolution of 256×256, and a spatial resolution of 0.3 meters. For consistency in this study, all HR images are reshaped to 256×256. We randomly select 75 images from each category for training and use the remaining 25 for testing. All models are trained for 100 epochs, and the batch size for Text-RSSR is set to 4.

The AID consists of 10,000 RGB images with a resolution of 600×600 pixels [41]. Introduced by G.-S. Xia *et al.*, it includes 30 scene categories, each containing several hundred images, with a spatial resolution of 0.5 meters. In this study, 80% of the images from each category are randomly selected for training and the remaining 20% for testing, yielding 8,000 training images and 2,000 testing images. To evaluate the models' adaptability to varying image resolutions, all HR images are resized to 576×576—differing from the settings used in other datasets—while LR images are resized to 144×144. All models are trained for 20 epochs, with the batch size for Text-RSSR set to 1.

The ILSVRC2012 dataset, introduced by O. Russakovsky *et al.* [42], is a widely used subset of ImageNet comprising 1,000 object categories, each with 1,300 images, along with a validation set of 50,000 images and a test set of 100,000

TABLE III
QUANTITATIVE RESULTS OF THE EXPERIMENT ON AID. THE BEST RESULT IS HIGHLIGHTED IN **BOLD**, WHILE THE SECOND-BEST RESULT IS UNDERLINED. THE UNIT FOR PSNR IS DECIBELS (DB).

Class	FAT [45]		FMSR [22]		GCRDN [46]		TSFNet [47]		TTST [34]		Text-RSSR	
	PSNR	**SSIM**	**PSNR**	**SSIM**	**PSNR**	**SSIM**	**PSNR**	**SSIM**	**PSNR**	**SSIM**	**PSNR**	**SSIM**
Airport	27.7525	0.7324	27.9582	**0.7419**	27.6506	0.7277	27.7524	0.7337	27.9100	0.7384	**27.9673**	0.7401
Bare Land	34.5639	0.8269	34.7455	0.8323	34.6666	0.8289	34.6324	0.8304	34.7469	0.8322	**34.8499**	**0.8328**
Baseball Field	27.7234	0.7363	**27.9125**	**0.7459**	27.6471	0.7350	27.7408	0.7401	27.7956	0.7427	27.9062	0.7454
Beach	31.5206	0.7959	31.7368	0.8017	31.6311	0.7967	31.6762	0.7993	31.7811	0.8025	**31.8627**	**0.8028**
Bridge	29.5164	0.8066	29.7611	**0.8140**	29.3744	0.8033	29.5317	0.8083	29.7168	0.8122	**29.7696**	0.8134
Center	24.1964	0.6563	24.4127	**0.6683**	24.0441	0.6476	24.2564	0.6598	24.3907	0.6659	**24.4493**	**0.6683**
Church	23.5623	0.6633	23.7468	**0.6747**	23.4374	0.6554	23.5907	0.6661	23.7075	0.6711	**23.7721**	0.6743
Commercial	24.3682	0.6655	24.5198	**0.6741**	24.2827	0.6595	24.4117	0.6687	24.4918	0.6717	**24.5252**	0.6737
Dense Residential	25.4483	0.6691	**25.5893**	**0.6774**	25.3104	0.6597	25.4380	0.6699	25.5417	0.6724	25.5836	0.6755
Desert	35.4974	0.8603	35.6769	0.8646	35.5656	0.8623	35.4717	0.8622	35.5750	0.8649	**35.8370**	**0.8652**
Farmland	32.3457	0.8001	**32.5888**	**0.8090**	32.2252	0.7972	32.2969	0.8011	32.4639	0.8054	32.5196	0.8064
Forest	25.6715	0.5934	25.8384	0.5994	25.6358	0.5859	25.7607	0.5953	25.8171	0.5971	**25.8607**	**0.5998**
Industrial	25.3502	0.6885	25.5426	**0.6982**	25.1632	0.6782	25.3566	0.6899	25.4803	0.6942	**25.5438**	0.6970
Meadow	30.5650	0.6421	30.7640	**0.6548**	30.6964	0.6493	30.7205	0.6529	30.7427	0.6536	**30.7743**	0.6544
Medium Residential	24.1298	0.5843	**24.3298**	0.5951	24.0050	0.5725	24.2174	0.5894	24.2783	0.5911	24.3296	**0.5952**
Mountain	26.9741	0.6309	27.0518	0.6337	26.9325	0.6250	26.9786	0.6326	27.0588	0.6346	**27.0870**	**0.6357**
Park	25.2941	0.6589	25.4161	0.6655	25.2218	0.6533	25.3075	0.6610	25.4103	0.6638	**25.4422**	**0.6663**
Parking	24.5978	0.7433	**24.9916**	**0.7607**	24.1571	0.7263	24.6874	0.7474	24.3829	0.7408	24.8294	0.7539
Playground	30.0128	0.7827	30.2509	**0.7918**	29.7851	0.7770	30.0155	0.7848	30.1723	0.7888	**30.2731**	0.7910
Pond	27.8899	0.7152	27.9814	0.7191	27.8211	0.7123	27.8758	0.7161	27.9796	0.7186	**28.0043**	**0.7200**
Port	26.2843	0.7471	26.4492	**0.7545**	26.1971	0.7440	26.2961	0.7489	26.4259	0.7528	**26.4616**	0.7541
Railway Station	25.5554	0.6620	25.7163	**0.6698**	25.3999	0.6506	25.5058	0.6609	25.6945	0.6667	**25.7389**	0.6695
Resort	22.7878	0.6255	22.9452	0.6342	22.7181	0.6206	22.7916	0.6278	22.9095	0.6325	**22.9499**	**0.6345**
River	28.5470	0.6859	28.6762	0.6926	28.5350	0.6848	28.5804	0.6897	28.6669	0.6919	**28.6964**	**0.6928**
School	24.0299	0.6776	24.2210	0.6871	23.9095	0.6716	24.0555	0.6810	24.1896	0.6849	**24.2368**	**0.6876**
Sparse Residential	24.6960	0.5454	24.8526	**0.5526**	24.6912	0.5397	24.7662	0.5477	24.8212	0.5498	**24.8562**	0.5524
Square	28.3506	0.7540	28.5661	**0.7636**	28.1843	0.7483	28.3606	0.7565	28.5292	0.7613	**28.5974**	0.7634
Stadium	26.0934	0.7377	26.3204	0.7486	25.9347	0.7310	26.1205	0.7399	26.2661	0.7471	**26.3626**	**0.7502**
Storage Tanks	24.8955	0.6788	**25.0844**	**0.6894**	24.7784	0.6727	24.9263	0.6815	25.0396	0.6863	25.0742	0.6881
Viaduct	25.6030	0.6395	25.7718	**0.6497**	25.4949	0.6314	25.6565	0.6427	25.7504	0.6467	**25.7897**	**0.6497**
Average	27.2062	0.7034	27.3927	**0.7120**	27.1123	0.6981	27.2373	0.7060	27.3347	0.7092	**27.4089**	0.7116

TABLE IV
QUANTITATIVE RESULTS OF THE EXPERIMENT ON ILSVRC2012 DATASET. THE BEST RESULT IS HIGHLIGHTED IN **BOLD**, WHILE THE SECOND-BEST RESULT IS UNDERLINED.

Model	**PSNR** (dB)	**SSIM**
FAT [45]	23.7829	0.6485
FMSR [22]	24.0107	0.6585
GCRDN [46]	23.9232	0.6527
TSFNet [47]	23.8665	0.6489
TTST [34]	24.0294	0.6591
Text-RSSR	**24.0528**	**0.6599**

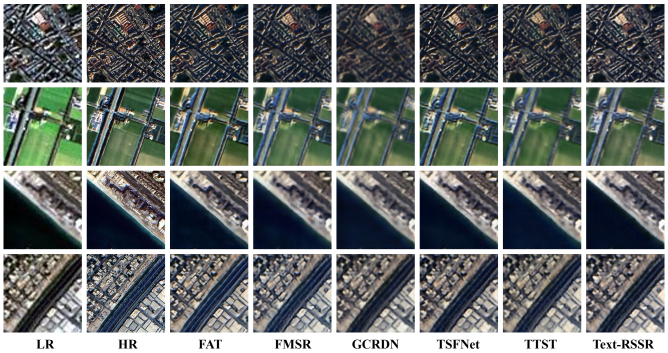
images. In this study, we construct the training set by randomly selecting 20 images from each category, resulting in a total of 20,000 images. For testing, 1,000 images are randomly sampled from the original ILSVRC2012 test set. All HR images are resized to 256×256, while the LR images are resized to 64×64. All models are trained for 20 epochs, with a batch size of 4 for Text-RSSR.

C. Performance Comparison

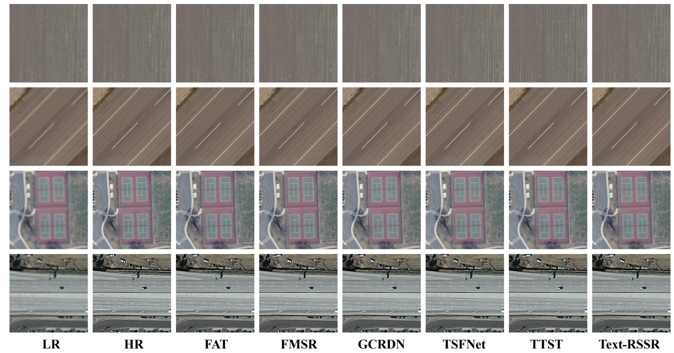
To evaluate the performance of the proposed Text-RSSR model, we compared the performance of Text-RSSR with other state-of-the-art methods from 2024 and 2025, including FAT [45], FMSR [22], GCRDN [46], TSFNet [47] and TTST [34]. This study employs two evaluation metrics to assess the quality of the SR images: PSNR and SSIM.

As shown in Tables I, II, III and IV, Text-RSSR achieves the highest PSNR and SSIM scores across all three datasets. The learnable parameter counts and FLOPs for each model are provided in Table I. These values are computed using Python's thop library, with FLOPs calculated for a 64×64 RGB input image. Compared with the TTST model, Text-RSSR improves PSNR by 0.0878 dB and SSIM by 0.0160, while increasing FLOPs by only 0.02 G. This highlights Text-RSSR's high computational efficiency. Due to the domain shift between LR and HR images in the Alsat-2B dataset, FAT and GCRDN exhibit a substantial drop in performance (PSNR $\leq$ 16 dB), indicating their limited ability to handle domain shift. Moreover, in the UC Merced Land Use Dataset and AID, which include a greater variety of image categories, the benefits of text guidance are more pronounced. In the UC Merced Land Use Dataset, containing 21 image categories, Text-RSSR achieves a PSNR of 27.4904 dB and an SSIM of 0.7775, surpassing the second-best results by 0.669 dB and 0.0037, respectively. In AID, which comprises 30 image categories, Text-RSSR attains a PSNR of 27.4089 dB and an SSIM of 0.7116, exceeding the second-best results by 0.0742 dB and 0.0024, respectively. These quantitative findings underscore the effectiveness of incorporating text guidance.

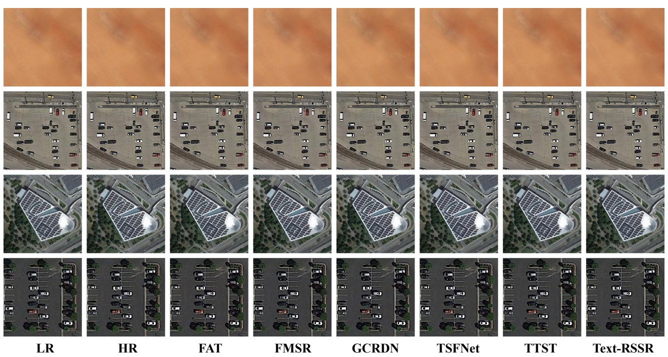
To further evaluate the visual quality of the SR images, we



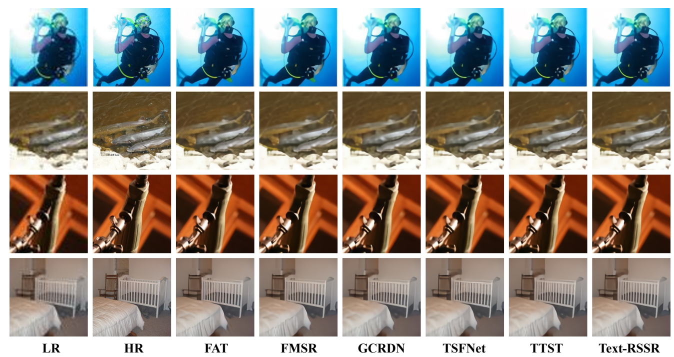
(a) Qualitative results on Alsat-2B dataset.



(b) Qualitative results on UC Merced Land Use Dataset.



(c) Qualitative results on AID.



(d) Qualitative results on ILSVRC2012 dataset.

Fig. 3. Visualization of results from Text-RSSR and other methods across all four datasets. Subfigures are labeled (a)–(d).

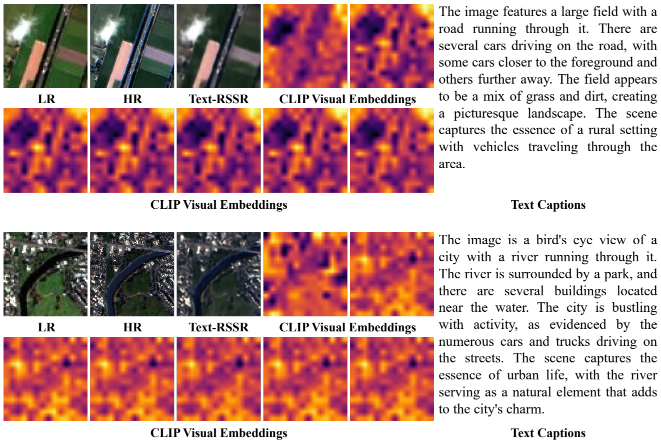
TABLE V

QUANTITATIVE RESULTS OF THE DOMAIN ADAPTATION EXPERIMENTS, WHERE "UC MERCED" DENOTES THE UC MERCED LAND USE DATASET. THE TRAINING DATASET IS LISTED TO THE LEFT OF THE →, AND THE TESTING DATASET TO THE RIGHT. THE BEST RESULT IS HIGHLIGHTED IN **BOLD**, WHILE THE SECOND-BEST RESULT IS <u>UNDERLINED</u>.

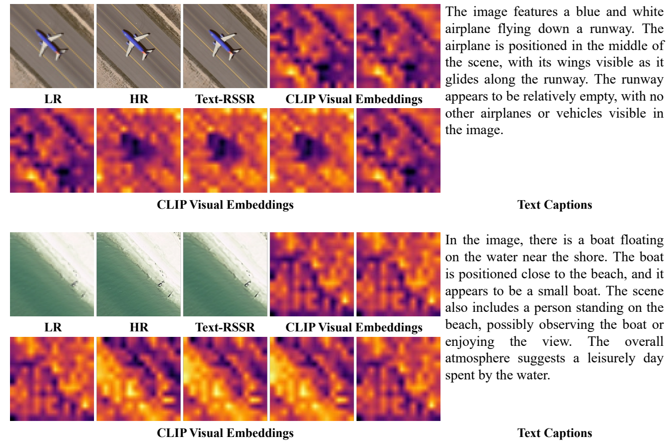
Model	Alsat-2B →UC Merced		UC Merced →Alsat-2B		ILSVRC2012 →Alsat-2B		ILSVRC2012 →UC Merced	
	PSNR (dB)	**SSIM**	**PSNR** (dB)	**SSIM**	**PSNR** (dB)	**SSIM**	**PSNR** (dB)	**SSIM**
FAT [45]	19.2788	0.4559	13.5311	0.2497	13.6382	0.2705	25.6201	0.6860
FMSR [22]	<u>19.9059</u>	0.5316	13.5659	0.2518	13.6966	0.2756	25.8744	<u>0.6938</u>
GCRDN [46]	**20.9480**	<u>0.5356</u>	13.5551	<u>0.2556</u>	<u>13.7073</u>	<u>0.2770</u>	<u>25.8940</u>	0.6934
TSFNet [47]	19.6710	0.4932	13.5017	0.2363	**13.7622**	**0.2782**	25.7703	0.6873
TTST [34]	19.2694	0.5094	**13.6064**	**0.2559**	13.6952	0.2754	25.7856	0.6906
Text-RSSR	19.7900	**0.5364**	<u>13.5755</u>	0.2552	13.6798	0.2753	**25.8997**	**0.6947**

compare the results of Text-RSSR with those of other models across all four datasets, as shown in Figure 3. In addition to achieving higher PSNR and SSIM scores, Text-RSSR also demonstrates superior visual fidelity. When a domain shift exists between LR and HR images, as illustrated in the first and second rows in Figure 3a, Text-RSSR successfully restores the correct color domain of the HR images. In cases involving repetitive line patterns, as seen in the first, second and fourth rows in Figure 3b, Text-RSSR reconstructs the lines directly rather than simply emphasizing them. Moreover, Text-RSSR performs particularly well on parking lot images, achieving state-of-the-art results compared with other methods, as shown in the second and fourth rows in Figure 3c. Lastly, Text-RSSR demonstrates superior performance in reconstructing common object images compared to other methods, as shown in Figure 3d. While it excels at recovering repeatable patterns, such as the railing of the baby bed in the last row, it also outperforms others in restoring contextual details, as seen in the reconstruction of the bell in the third row.

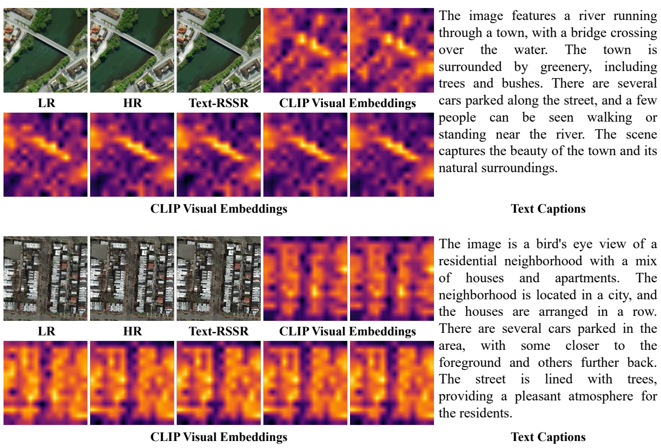
We also present a visualization of Text-RSSR results across all four datasets in Figure 4, which includes both the text captions and heatmap visualizations of the CLIP Visual Embeddings. The text captions often provide additional details that may be overlooked by the CLIP Visual Embeddings, thereby contributing missing information during the image super-resolution process. For instance, in the first sample in Figure 4c, the text caption references trees and bushes, which are not captured by the CLIP Visual Embeddings. The figure further illustrates that the model exhibits varying focus at different stages of the Text-Guided Iterative Super-Resolution process. For example, in the second sample of Figure 4b, the model emphasizes the sea during the third, fourth, and fifth



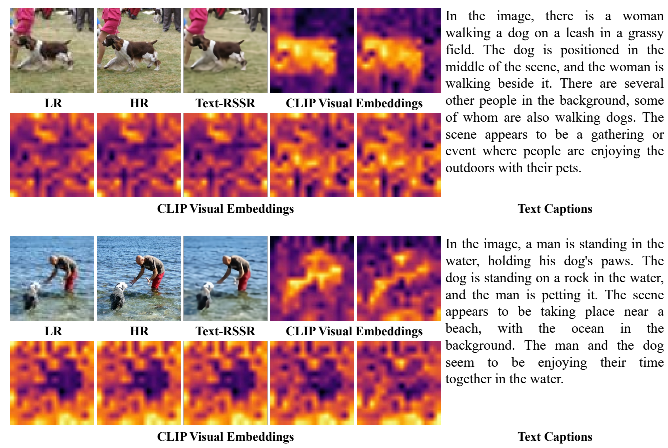
(a) Qualitative results on Alsat-2B dataset.



(b) Qualitative results on UC Merced Land Use Dataset.



(c) Qualitative results on AID.



(d) Qualitative results on ILSVRC2012 dataset.

Fig. 4. Visualization of Text-RSSR results across all four datasets, including both the text captions and heatmap visualizations of the CLIP Visual Embeddings. The text captions often provide additional details that may be overlooked by the CLIP Visual Embeddings, while the CLIP Visual Embeddings from different iterations in Text-Guided Iterative Super-Resolution often have different attention on the image. Subfigures are labeled (a)–(d).

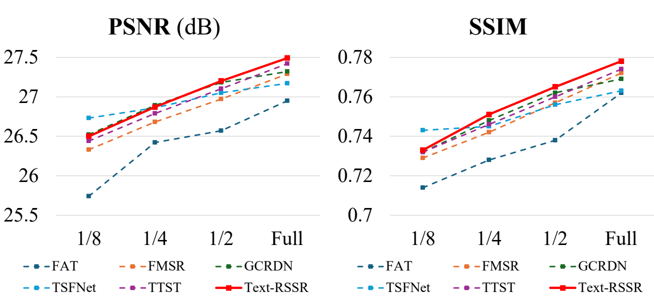


Fig. 5. Visualization of the data scalability experiment results on the UC Merced Land Use Dataset.

iterations, while focusing on the beach and high-frequency contextual details in other iterations. In summary, both the text captions generated by the VLM and the corresponding CLIP Visual Embeddings contribute meaningfully to HR image reconstruction.

To further evaluate the domain adaptability of Text-RSSR, we conduct a domain adaptation experiment between the Alsat-2B dataset and the UC Merced Land Use Dataset. Additionally, we perform another experiment in which models are trained on the ILSVRC2012 dataset and tested on both the Alsat-2B and UC Merced Land Use datasets. The quantitative results are summarized in Table V. These results indicate that Text-RSSR exhibits strong domain adaptability. Specifically, it achieves the highest SSIM index (0.5364) when trained on the Alsat-2B dataset and tested on the UC Merced Land Use Dataset. It also attains the second-best PSNR value (13.5755 dB) when trained on the UC Merced dataset and tested on Alsat-2B. Moreover, when trained on common object images, Text-RSSR achieves the best performance in both PSNR and SSIM when evaluated on the UC Merced dataset, which contains a broader variety of scene types than the Alsat-2B dataset. Overall, these findings confirm that Text-RSSR is highly adaptable across domains, and its performance can be further enhanced by training on datasets with greater scene diversity.

To further assess the scalability of Text-RSSR under limited training data, we conduct a data scalability experiment on the UC Merced Land Use Dataset. Specifically, we construct

TABLE VI
QUANTITATIVE RESULTS OF THE DATA SCALABILITY EXPERIMENTS ON THE UC MERCED LAND USE DATASET. "1/8", "1/4", "1/2", AND "FULL" INDICATE THE PROPORTION OF THE TRAINING SET USED. THE BEST RESULT IS HIGHLIGHTED IN **BOLD**, WHILE THE SECOND-BEST RESULT IS UNDERLINED.

Model	1/8		1/4		1/2		Full	
	PSNR (dB)	**SSIM**	**PSNR** (dB)	**SSIM**	**PSNR** (dB)	**SSIM**	**PSNR** (dB)	**SSIM**
FAT [45]	25.7443	0.7137	26.4231	0.7280	26.5719	0.7376	26.9460	0.7617
FMSR [22]	26.3315	0.7290	26.6758	0.7418	26.9741	0.7571	27.2922	0.7718
GCRDN [46]	<u>26.5177</u>	0.7319	**26.8927**	<u>0.7484</u>	<u>27.1808</u>	<u>0.7620</u>	27.3259	0.7688
TSFNet [47]	**26.7270**	**0.7428**	26.8622	0.7475	27.0533	0.7563	27.1664	0.7631
TTST [34]	26.4420	0.7324	26.7914	0.7461	27.1010	0.7603	<u>27.4235</u>	<u>0.7738</u>
Text-RSSR	26.5027	<u>0.7331</u>	<u>26.8675</u>	**0.7512**	**27.1958**	**0.7648**	**27.4904**	**0.7775**

TABLE VII
QUANTITATIVE RESULTS OF THE ABLATION STUDY ON TEXT-GUIDED FEATURE ALIGNING FOR THE UC MERCED LAND USE DATASET. *CLIP V. E.* REFERS TO CLIP VISUAL EMBEDDINGS, *T. E.* TO TEXTUAL EMBEDDINGS, *Hierarchy* TO THE HIERARCHICAL ARCHITECTURE OF THE TEXT-GUIDED FEATURE ALIGNING PROCESS, AND *A. F.* TO THE LEARNABLE ALIGNING FACTOR. THE BEST RESULT IS SHOWN IN **BOLD**, WHILE THE SECOND-BEST RESULT IS UNDERLINED.

Model Settings				Metrics	
CLIP V. E.	T. E.	Hierarchy	A. F.	**PSNR** (dB)	**SSIM**
✓				27.4464	0.7726
	✓			27.4506	0.7744
✓	✓			27.4580	0.7750
✓	✓	✓		<u>27.4771</u>	<u>0.7773</u>
✓	✓	✓	✓	**27.4904**	**0.7775**

three reduced training sets containing only 1/2, 1/4, and 1/8 of the original training images, respectively. The quantitative results are shown in Table VI, and the visualization of the results is shown in Figure 5. The results indicate that Text-RSSR exhibits the fastest performance growth as the training set size increases, while still maintaining competitive results with very limited data. Specifically, Text-RSSR achieves state-of-the-art performance when trained on half and the full partition of the training data. It also attains the second-best SSIM when trained on 1/8 of the data, and ranks second in PSNR and first in SSIM when trained on 1/4 of the data. This suggests that the incorporation of text guidance not only improves reconstruction quality with full training data but also provides strong robustness in low-data regimes, highlighting the model's generalization capability and practical utility in scenarios where annotated data is scarce.

D. Ablation Study

To further evaluate the effectiveness of each components of Text-Guided Feature Aligning process, this study conducts an ablation study on Text-Guided Feature Aligning on the UC Merced Land Use Dataset. We first construct a model with a single Text-Guided Feature Aligning process applied before the first SR Unit, using only CLIP Visual Embeddings and removing the Textual Embeddings. Next, we build a second model that uses only Textual Embeddings, excluding the CLIP Visual Embeddings. We then create a third model that incorporates both CLIP Visual Embeddings and Textual Embeddings. Subsequently, we design a hierarchical architecture by adding a Text-Guided Feature Aligning process after each SR Unit. Finally, we introduce a learnable aligning factor for each Text-Guided Feature Aligning process to fine-tune the alignment, resulting in the complete Text-RSSR model.

The quantitative results of the ablation study are presented in Table VII. As shown, incorporating CLIP Visual Embeddings as pixel-wise guidance or Textual Embeddings as channel-wise guidance both enhance performance, with Textual Embeddings yielding a slightly greater improvement (0.0042 dB in PSNR and 0.0018 in SSIM) compared to CLIP Visual Embeddings. Combining both types of embeddings further boosts reconstruction quality, demonstrating that pixel-wise and channel-wise guidance provide complementary information for the feature aligning process. Introducing the hierarchical architecture brings an additional performance gain (0.0191 dB in PSNR and 0.0023 in SSIM), indicating that multi-level alignment enhances the integration of visual and textual cues. Finally, adding a learnable aligning factor provides the Text-RSSR model with greater flexibility in leveraging text guidance, leading to a further improvement in performance.

V. CONCLUSION

This study proposes Text-RSSR, a novel framework for leveraging text guidance in remote sensing image super-resolution. By integrating multimodal cues through CLIP embeddings and iteratively aligning visual and textual features, Text-RSSR effectively enhances the reconstruction of fine-grained spatial details. The four-step pipeline (Data Preparation, Multimodal Encoding, Text-Guided Iterative Super-Resolution, and Image Decoding) demonstrates how semantic information from text captions can be systematically incorporated into the super-resolution process. This approach highlights the potential of multimodal learning to bridge the gap between low-resolution and high-resolution imagery, offering a promising direction for advancing remote sensing applications that demand both high fidelity and semantic consistency.

The experimental results further validate the effectiveness of the proposed Text-RSSR framework. Across all four benchmark datasets: Alsat-2B, UC Merced Land Use, AID, and ILSVRC2012, Text-RSSR consistently outperforms state-of-the-art methods in both PSNR and SSIM, while maintaining competitive computational efficiency. Notably, Text-RSSR achieves substantial improvements in scenarios with pronounced domain shifts and complex scene variations, underscoring the robustness of its text-guided alignment mechanism. In the UC Merced and AID datasets, which cover diverse scene categories, Text-RSSR demonstrates particularly strong gains, highlighting the value of multimodal guidance in handling heterogeneity in remote sensing imagery. Moreover, qualitative evaluations reveal that Text-RSSR excels in restoring fine structural details and semantic consistency, such as reconstructing repetitive line patterns and preserving correct color domains under challenging conditions. The ablation study further confirms that the hierarchical integration of CLIP Visual and Textual Embeddings, together with a learnable aligning factor, is critical to achieving optimal performance. Collectively, these findings establish Text-RSSR as a highly effective and efficient framework for advancing the state of remote sensing image super-resolution.

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